

MADHURA DOLE

Bellevue, WA | madhuradole@gmail.com | +1-(214)-729-3832 | LinkedIn: [madhuradole](#)

PROFESSIONAL SUMMARY

Software Engineer with 8+ years of experience designing, building, and supporting backend systems and critical software platforms. Strong background in backend engineering, system design, technical leadership, observability, and production reliability across Python, Java, C#/.NET, AWS, Kafka, Kubernetes, and SQL databases. Proven track record of delivering regulated and high-impact projects, improving operational efficiency, and building APIs, automation, and monitoring solutions.

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, JavaScript

Backend / Frameworks: REST APIs, Spring Boot, .NET, Workflow Orchestration, Microservices

Frontend: React, Vue.js, TypeScript

Databases: PostgreSQL, MySQL, SQLite, Oracle, Snowflake, Relational Schema Design

Cloud / Infra: AWS (EC2, ECS, S3, SQS, Lambda, Glue), Terraform

Observability / Operations: Splunk, Grafana, Monitoring, Alerting, Production Support, Root Cause Analysis

Engineering Practices: Automated Testing, Code Quality, CI/CD, Debugging, Agile

PROFESSIONAL EXPERIENCE

Software Engineer II | Chewy Inc.

Nov 2022 – Present

Tech Stack: Python, Postgres, SQLite, AWS, EC2, Lambda, REST APIs, Java, Spring Boot, Splunk, React, TypeScript, Vue

- **Leading the technical architecture and end-to-end implementation of Curated Insight Assistant**, an internal full-stack analytics workflow application that enables workflow authoring, SQL-based data retrieval from Snowflake, AI-generated insights and narratives, scheduled execution through Airflow, and email alert delivery.
- **Designed architecture for a workflow orchestration framework** to trigger and monitor simulations via REST APIs, track progress in real-time, and support scheduled simulation runs using cron-based scheduling. Completed requirements gathering, technical design, and flyway infrastructure setup for Snowflake database DDL updates.
- Implemented **Data Modifiers** in the Order Routing System Simulations (Python) project, **resulting in an 80% reduction in the manual effort** required to modify system data as per the simulation request.
- Designed and built an **MPS Comparator Tool** to **identify fidelity gaps between production and simulation optimizer input files**.
- Created a comprehensive **monitoring and alerting framework** in Splunk for improved observability and faster defect resolution.
- Integrated data science-built optimizer modules into the Order Routing System Simulations infrastructure.
- **Delivered multiple front-end features using Vue.js**, including S3 file upload support and JSON simulation configuration support.

Senior Software Engineer | VISA Inc.

Mar 2020 – Sept 2022

Tech Stack: .NET, C#, REST APIs, Java, MySQL, MS-SQL, DB2, Spring Boot, Kafka, Kubernetes, Grafana, Splunk

- **Led a technical project with a team of 2 developers for the Payment Controls module** in the Digital Wallet project, to synchronize Payment Controls between a Pseudo Account behind the Virtual Account and a Digital Wallet. Tech stack: **.NET/C#, MS-SQL, Kafka, Kubernetes**. **Roles and responsibilities:** Design architect, development lead, QA, production release, support.
- **Built 4 high-performance .NET REST APIs (200–400ms average latency) and Kafka consumers and producers in C#** to process transaction notification messages; built Grafana Dashboard for Data Visualization and setup critical, actionable PromQL alerts.
- **Spearheaded the India Data Localization project** for Visa Payment Controls; developed .NET APIs for data storage compliance to store transaction data of India-based issuers in India data center.
- Implemented a security feature for invoice file cleaning in **Visa Payables Automation** using Visa's file sanitization APIs (Java, Spring Boot, MySQL).
- Built Grafana dashboards and Splunk alerts for real-time operational monitoring across key systems.

Product Reliability Engineer | VISA Inc.

Jun 2018 – Feb 2020

Tech Stack: .NET, C#, MS-SQL, WCF, IBM MQ/MSMQ, Windows/Linux

- Designed and built a Windows application (C#, SQL) to update Visa customer card statuses, with alert-triggering mechanism on each update. **Reduced incident resolution time by 80%, eliminating manual DML operations.**
- Created tools to automate support tasks, increasing team productivity by **up to 85%**.
- Handled vulnerability remediation in a **24x7 production environment** and led multiple production deployment efforts for Visa's Loyalty and Marketing system.

R&D Programmer | Center for Modelling & Simulation (UT Dallas)

Aug 2017 – May 2018

Tech Stack: ML/NLP – Unity, C#, MS HoloLens, Python

- **Co-developed a question-answering, augmented reality-based interface** using Unity and C#, simulating realistic human-machine interactions.
- Co-developed a Recurrent Neural Network composed of POS tagger, named entity recognizer, and sentiment analyzer.
- Implemented a POS Tagger using Hidden Markov Models and rule-based approach with 92.67% accuracy.

Software Engineer Intern | VISA Inc.

May – Aug 2017

Tech Stack: Java, J2EE, JSP, MySQL, MS SQL, HTML, CSS, JavaScript

- Developed a Visa internal website to create a Production Server Certificate Dashboard UI.

Senior Software Developer | TATA Consultancy Services

Jul 2013 – Jun 2016

Tech Stack: C#, .NET, MS-SQL, WCF, VB.NET

- **Developed a C# language interpreter** for OTX (ISO specified XML files).
- **Led a client project module**, serving as the offshore point of contact and mentoring new employees.
- Built several Windows apps for process improvement using WCF, VB.NET and MS-SQL.
- **Led a 6-member team** to build a hotel management website using Java, JDBC, JSP, Servlets, and MVC architecture.
- Completed various actions items for database design, including but not limited to: ER modeling, SQL scripts, triggers, procedures, and constraints.

EDUCATION

M.S. Computer Science, The University of Texas at Dallas (Concentration: Intelligent Systems [AI/ML]) (GPA: 3.5) May 2018

Relevant Courses: Advanced Machine Learning, Artificial Intelligence, Natural Language Processing.

B.E. in Information Technology, Dr. Babasaheb Ambedkar Marathwada University, India (GPA: 4.0[WES]) May 2013

Relevant Courses: Data Structures & Algorithms, Programming in Java, Object Oriented Programming Systems

AWARDS & CERTIFICATES

- Above and Beyond Silver Award – Leadership and Collaboration, Visa (Aug 2021)
- Above and Beyond Silver Award – Innovation (Pioneering with a Purpose), Visa (Jan 2019)
- Certificate of Completion – Advanced Java and Spring Boot (Oct 2020)
- Oracle Certified Associate, Java SE 7 Programmer (Mar 2015)